

FR350SDF Rear Helper Spring Kit Installation Manual

Model: FR350SDF (Rear Suspension Package)

Please read these instructions carefully before commencing installation. Ensure all standard safety protocols are observed when working beneath the vehicle.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Hex Cap Screw (HCS) (3/8" x 1" UNF, Grade 5, Zinc Plated)	2
2	Split Lock Washer (3/8", Grade 5, Zinc Plated)	4
3	Flat Washer USS (3/8" x 1", Grade 2, Zinc Plated)	2
4	Weldment Bracket (For Left-Hand & Right-Hand installation)	2
5	Rubber Spacer (4" O.D. x 1/2" Thick with Bushing)	2
6	Hollow Rubber Spring	2
7	Hex Cap Screw (HCS) (3/8" x 2" UNF, Grade 5, Zinc Plated)	2
8	Hex Cap Screw (HCS) (3/8" x 1-1/2" UNF, Grade 5, Zinc Plated)	2
9	Nylock Locknut (3/8" UNF, Grade 5, Zinc Plated)	2

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate lifting device, allowing the rear axle to hang freely. Secure the chassis with stands before proceeding.
- 2. Remove OEM Components:** Detach and remove the existing original equipment manufacturer (OEM) bump stop and its associated bushing from the vehicle frame.
- 3. Assemble Helper Spring:** Combine the hollow rubber spring (Ref 6) with the weldment bracket (Ref 4). Refer to diagram C (if provided in original source) for correct bracket orientation.
- 4. Mount Assembly:** Install the assembled helper spring unit onto the vehicle frame utilizing the existing mounting holes and the provided fasteners (Ref 1, 2, 3, 7, 8, 9).
- 5. Lower Vehicle:** Carefully lower the vehicle back to the ground.

Installation Notes:

- The rubber spacer (Ref 5) and the 3/8" x 2" UNF hex cap screw (Ref 7) can be utilized to achieve a specific desired ride height.
- Verify that brake lines or cables do not interfere with the helper spring's position or movement.
- When necessary to center the helper spring over the axle, use the 3/8" UNF Nylock locknut (Ref 9) with the offset holes on the bracket.
- If the original equipment vehicle features a peg-style casting at the bump stop location, the helper spring is designed to make contact with this peg. Do not remove the OEM peg.

3. Clearance Instructions & Fitting Guidance

- Clearance Gap: For rear helper spring kits installed on pickup trucks and vans, an ideal clearance of 1/2" to 1-1/2" should be maintained between the rubber spring and its contact point when the vehicle is unladen.
- Load Effects: The presence of a partial load, cargo, or aftermarket vehicle accessories will reduce or eliminate this initial clearance gap.
- Heavy-Duty Applications: For medium and heavy-duty trucks, trailers, and motorhomes, the helper springs are designed for constant contact with no gap.

Important Safety Warnings:

- Never exceed the vehicle manufacturer's specified gross axle weight rating (GAWR).
- Inspect for any potential interference between the hollow rubber spring and other vehicle components. The rubber spring's diameter will expand under load.

Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.